

ML PRACTICAL 3

NAME : ISHA BHANGRE

SECTION-BATCH : B-B2

ROLL NO. : 20

```
In [4]: import pandas as pd

data = pd.read_csv("enjoy.csv")
data
```

```
Out[4]:
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	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
0	Sunny	Warm	Normal	Strong	Warm	Same	Yes
1	Sunny	Warm	High	Strong	Warm	Same	Yes
2	Rainy	Cold	High	Strong	Warm	Change	No
3	Sunny	Warm	High	Strong	Cool	Change	Yes
4	Sunny	Cold	Normal	Weak	Warm	Same	No
5	Cloudy	Warm	Normal	Weak	Warm	Same	No
6	Sunny	Warm	Normal	Weak	Cool	Same	Yes
7	Rainy	Warm	High	Weak	Cool	Change	No
8	Sunny	Warm	Normal	Strong	Cool	Same	Yes
9	Sunny	Warm	High	Weak	Warm	Same	Yes

```
In [5]: X = data.iloc[:, :-1].values

y = data.iloc[:, -1].values
```

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In [10]: import numpy as np

def find_s(X,y):
    hypothesis = None

    print("Initial Hypothesis: ", hypothesis)
    for i in range(len(X)):
        if y[i] == "Yes":
            if hypothesis is None:
                hypothesis = X[i].copy()
            else:
                for j in range(len(hypothesis)):
                    if hypothesis[j] != X[i][j]:
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        hypothesis[j] = "?"
        print(f"\nAfter Training Example {i+1}")
        print(hypothesis)
    return hypothesis

final_hypothesis = find_s(X,y)

print("\nFinal Hypothesis")

print(final_hypothesis)

```

Initial Hypothesis: None

After Training Example 1

['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']

After Training Example 2

['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']

After Training Example 3

['?' '?' '?' 'Strong' 'Warm' '?']

After Training Example 4

['?' '?' '?' 'Strong' 'Warm' '?']

After Training Example 5

['?' '?' '?' '?' 'Warm' '?']

After Training Example 6

['?' '?' '?' '?' 'Warm' '?']

After Training Example 7

['?' '?' '?' '?' 'Warm' '?']

After Training Example 8

['?' '?' '?' '?' '?' '?']

After Training Example 9

['?' '?' '?' '?' '?' '?']

After Training Example 10

['?' '?' '?' '?' '?' '?']

Final Hypothesis

['?' '?' '?' '?' '?' '?']

```

In [12]: def candidate_elimination(concepts, target):
        specific = concepts[0].copy()
        general = [["?" for i in range(len(specific))]]
        print("\nInitial S:",specific)
        print("Initial G:",general)
        for i, h in enumerate(concepts):
            if target[i] == "Yes":
                for x in range(len(specific)):
                    if h[x] != specific[x]:
                        specific[x] = "?"

```

```
        else:
            for x in range(len(specific)):
                if h[x] != specific[x]:
                    general.append(["?" if j != x else specific[x] for j in range(1
print("\nExample",i+1)
print("S =", specific)
print("G =", general)
    return specific,general
S,G = candidate_elimination(X,y)

print("\nFinal Specific Boundary")

print(S)

print("\nFinal General Boundary")

for g in G:
    print(g)
```

```
Example 8
S = ['Sunny' 'Warm' '?' '?' '?' '?']
G = [['?', '?', '?', '?', '?', '?'], ['Sunny', '?', '?', '?', '?', '?'], ['?', 'War
m', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?',
'Same'], ['?', 'Warm', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?',
```

```
['?', '?', 'Strong', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?'], ['Sunny', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', 'Strong', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['Sunny', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?']]
```

Example 9

```
S = ['Sunny' 'Warm' '?' '?' '?' '?']
G = [['?', '?', '?', '?', '?', '?'], ['Sunny', '?', '?', '?', '?', '?'], ['?', 'War
m', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?',
'Same'], ['?', 'Warm', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?',
 '?', '?', 'Strong', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?',
 '?', '?'], ['?', '?'], ['Sunny', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?',
 '?'], ['?', '?', '?', 'Strong', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?',
 '?', '?', '?', '?'], ['?', '?'], ['Sunny', '?', '?', '?', '?', '?'], ['?', '?', '?',
 '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?',
 '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?',
 '?'], ['?']]
```

Example 10

```
S = ['Sunny' 'Warm' '?' '?' '?' '?']
G = [['?', '?', '?', '?', '?', '?'], ['Sunny', '?', '?', '?', '?', '?'], ['?', 'War
m', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?',
'Same'], ['?', 'Warm', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?',
 '?', '?', 'Strong', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?',
 '?', '?'], ['Sunny', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?',
 '?', '?', 'Strong', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?',
 '?', '?'], ['Sunny', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?',
 '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?',
 '?']]
```

Final Specific Boundary

```
['Sunny' 'Warm' '?' '?' '?' '?']
```

Final General Boundary

```
[ '?', '?', '?', '?', '?', '?']
['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', 'Same']
['?', 'Warm', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', 'Strong', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['Sunny', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', 'Strong', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['Sunny', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
```

In []: